



Distributed Systems

1st Lecture

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Outline

- Introduction to Distributed Systems
- Clock
- Q&A



End-to-End Arguments [Saltzer84]

■ Systems Design Principle

: Features like error recovery, security, and reliability should be built into application endpoints

- Endpoint Advantage Due to Complete Context and Data Awareness
- Low-Level Limitation Due to Partial Context and No Guarantee of Correctness
- Redundancy Due to Redundant Low-level Effort



End-to-End Arguments [Saltzer84] (Cont'd)

■ Practical Examples

□ File Transfers

- Network-layer prevention of packet loss
- End-to-end checksum in a file transfer app

□ Encryption

- Encrypting data end-to-end
- Network-layer encryption

□ Duplicate Suppression

- Duplicate messages through lower network layers
- End-to-end duplicate suppression

Distributed Systems

- Systems Where Components of Networked Computers Communicate and Coordinate Their Actions via Message Passing

- ☐ Concurrency

- ☐ No Global Clock

- ☐ Independent Failures

- Sharing States
- Providing Services
- Having Global Properties
- Heterogeneity
- Unreliable, Insecure, Costly Comm.
- Scalability
- Autonomy

- Systems Consisting of Collections of Spatially Separated Processes Communicating by Exchanging Messages

Different-Scale Systems

■ Systems of Increasing Scale & Decreasing Integration

System Type	Heterogeneity, Geographic Distribution	Lack of Centralized Control	Exemplar Computational Model
End (Single) System			Multithreading
Cluster			Distributed Shared Memory
Intranet	○		Manager/Worker
Internet	○	○	Collaborative Systems



Example of Distributed Systems

Figure 1.3 on Page 25



Distributed System Goals

■ Accessibility

- As Connectivity and Sharing Increases, Security and Privacy Matter

■ Distribution Transparency

- Access: Regarding Data Representation & Access Method
- Location: Regarding Resource Location
- Migration: Regarding Resource Movement
- Relocation: Regarding Movement in Use
- Replication: Regarding Resource Replication



Distributed System Goals (Cont'd)

- Distribution Transparency (Cont'd)

- Concurrency: Regarding Competitive Sharing
- Failure: Regarding Resource Failure and Recovery

- Openness:

Following Standard Rules That Describe the Syntax and Semantics of Services

- Interoperability

Distributed System Goals (Cont'd)

- Openness (Cont'd)

- Portability

- Extensibility

Separating Policy from Mechanism

- Scalability

- Limitations

- Centralized services

- Centralized data

- Centralized algorithms

- Synchronous communication



Distributed System Goals (Cont'd)

■ Scalability (Cont'd)

□ Decentralized Characteristics

- No local maintenance of global system state
- Decision making based only on local information
- Localized Failure
- No global clock

□ Scaling Techniques

- Asynchronous communication
- Client and Server load balancing
- Distribution: e.g., DNS & WWW
- Replication & caching

Distributed System Goals (Cont'd)

■ Performance

- Network latency
- Network bandwidth
- Communication overhead
- Synchronization
 - Locks, barriers, consensus, and coordination
- Load imbalance
- Data locality
- Contention
- Tail latency
- Fault handling
- Scalability

Distributed System Goals (Cont'd)

■ Comprehensibility

: How easy the system is to understand, reason about, debug, and predict

- Concurrency
- Configuration complexity
- Partial failure
- Dynamic behavior
- Asynchrony
- Non-determinism
- Complex interaction
- Emergent behavior
- Poor observability

[GPT-5.6 Luna]

Distributed System Goals (Cont'd)

■ Performance vs Comprehensibility

□ A Highly Optimized System

- Asynchronous execution
- Aggressive caching
- Batching
- Speculative execution
- Dynamic scheduling
- Replication
- Adaptive algorithm
- Multiple levels of parallelism

Distributed System Goals (Cont'd)

■ Performance vs Comprehensibility (Cont'd)

□ Simple / Comprehensible

- Easier debugging and prediction
- Potentially lower performance

□ Highly Optimized / Complex

- Potentially better performance and scalability
- Harder debugging, prediction, and performance attribution



Distributed System Goals (Cont'd)

■ Pitfalls

- Network Reliability
- Network Security
- Network Homogeneity
- Static Topology
- Zero Latency
- Infinite Bandwidth
- Zero Transport Cost
- One Administrator



Technical Goals

- Heterogeneity
 - H/W, S/W, and Data Components
- Varying Component Size and Extent
- Network Connection
- Uniform Set of Services
- Certain Global Properties



Distributed Computing Approaches

- Grid
 - Addressing Infrastructure
- Peer-to-Peer
 - Addressing Failure
 - Self-organizing into network topologies
 - w/o a Global Server or Authority



Distributed System Types

■ Distributed Computing Systems

□ Cluster Computing Systems

- Collection of computers connected in a high-speed network

□ Grid Computing Systems

- Federation of computer systems possibly in different administrative domains

□ Cloud Computing Systems

■ Distributed Information Systems

□ Transaction Processing Systems

- Atomic, Consistent, Isolated, and Durable Transaction



Distributed System Types (Cont'd)

■ Distributed Information Systems (Cont'd)

□ Enterprise Application Integration

■ Communicating Independent Components

- Remote procedure call
- Remote method invocation
- Message-oriented middleware w/ logical contact points
 - Message queuing model
 - Message brokers as application-level gateway w/ subscription & publication

□ Distributed Pervasive Systems

- Home Systems Possibly w/ UPnP
- Electronic Health Care Systems in a BAN
- Sensor Networks

Distributed System Types (Cont'd)

- Distributed Information Systems (DIS)
 - : Data, computation, applications, and/or services are distributed, providing a service
 - Distributed Databases
 - Distributed File/Storage Systems
 - Web-scale Information Systems
 - Enterprise Information Systems
 - IoT Information Systems
 - Distributed AI Systems



Issues

- Problem

- No Global Clock

- Issues

- How to Determine an Order of Events

- How to Determine Global States

- Consistency

Clock

■ Skew

- Difference between the Readings of Any Two Clocks

Figure 14.1 on Page 614

Coulouris, Dollimore and Kindberg Distributed Systems: Concepts and Design Edn. 5 © Pearson Education 2012

■ Drift

- Divergence of Clocks due to Counting Times at Different Rates

UTC (Coordinated Universal Time)

- International Time Standard

- Formerly, Greenwich Mean Time or GMT

- Zero hours UTC: midnight Greenwich (0 degrees longitude)

- Based on Atomic Time (Drift Rate: $1/10^{13}$ Seconds/Second)

- Signals synchronized and broadcast regularly

- From land-based radio stations and satellites



Synchronous vs Asynchronous

■ Synchronous Systems

□ Known Bounds

- Drift rate of clocks
- Max message transmission delay
- Time to execute each step of a process

■ Asynchronous Systems

□ No Bounds